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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 30th June Morning Shift

JEE Main Question Paper with Answer Key

30 Jun 2022	90	24	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	19 questions
Physics	25 questions
Out of syllabus	24 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

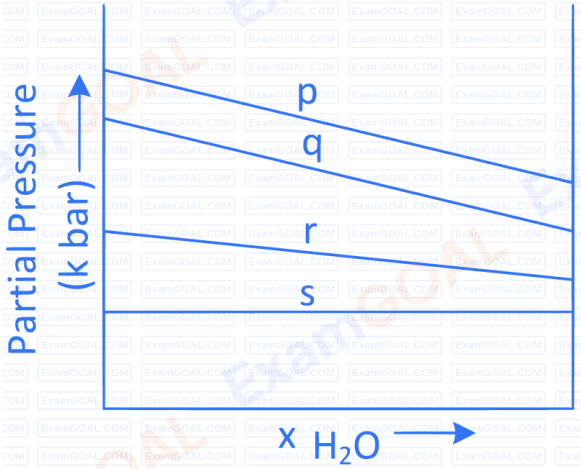
The number of radial nodes and total number of nodes in 4p orbital respectively are :

(A) 2 and 3 (B) 2 and 2 (C) 3 and 4 (D) 4 and 4

Open solution

Q2. MCQ (Single Correct Answer)

For a solution of the gases A, B, C and D in water at 298 K, the values of Henry's law constant (K_H) are 30.40, 2.34, 1.56×10^{-5} and 0.513 k bar respectively. In the given graph, the lines marked as 'p' and 's' correspond respectively to :



(A) A and C (B) B and A (C) D and A (D) C and D

Open solution

Q3. MCQ (Single Correct Answer)

The equilibrium constant for the reversible reaction

$2A(g) \rightleftharpoons 2B(g) + C(g)$ is K_1

$\frac{3}{2}A(g) \rightleftharpoons \frac{3}{2}B(g) + \frac{3}{4}C(g)$ is K_2.

K_1 and K_2 are related as :

(A) $K_1 = \sqrt{K_2}$ (B) $K_2 = \sqrt{K_1}$ (C) $K_2 = K_1^{3/4}$ (D) $K_1 = K_2^{3/4}$

Open solution

Q4. MCQ (Single Correct Answer)

In which of the following half cells, electrochemical reaction is pH dependent?

(A) $Pt \,|\, Fe^{3+}, Fe^{2+}$ (B) $MnO_4^- \,|\, Mn^{2+}$ (C) $Ag \,|\, AgCl \,|\, Cl^{-1}$ (D) $\frac{1}{2}F_2 \,|\, F^-$

Open solution

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Q5. MCQ (Single Correct Answer)

The correct order of electron gain enthalpy (– ve value) is :

(A) O > S > Se > Te (B) O < S < Se < Te (C) O < S > Se > Te (D) O < S > Se < Te

Open solution

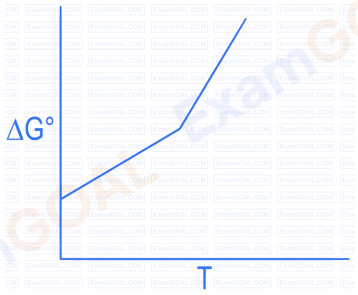
Q6. MCQ (Single Correct Answer)

ΔG° vs T plot for the formation of MgO, involving reaction

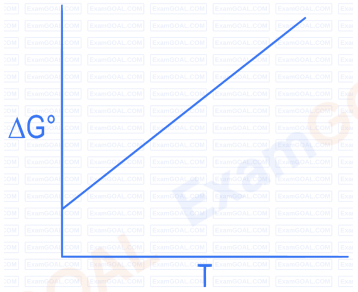
$2Mg + O_2 \rightarrow 2MgO$, will look like :

Open solution

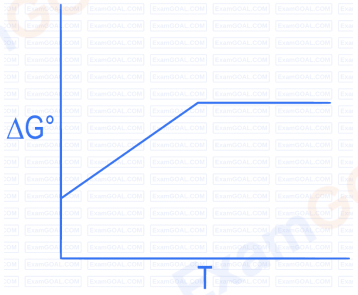
(A)



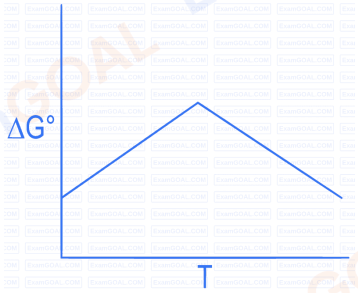
(B)



(C)



(D)



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Q7. MCQ (Single Correct Answer)



Open solution

A $\xrightarrow{573\ K}$ Red phosphorus $\xrightarrow[under\ pressure]{heat\ ;\ 803\ K}$ B

Red phosphorus is obtained by heating "A" at 573 K, and can be converted to "B" by heating at 803 K under pressure.

A and B, respectively, are

(A) β-black phosphorus and white phosphorus.

(B) white phosphorus and β-black phosphorus.

(C) α-black phosphorus and white phosphorus.

(D) white phosphorus and α-black phosphorus.

Q8. MCQ (Single Correct Answer)



Open solution

Correct formula of the compound which gives a white precipitate with $BaCl_2$ solution, but not with $AgNO_3$ solution, is :

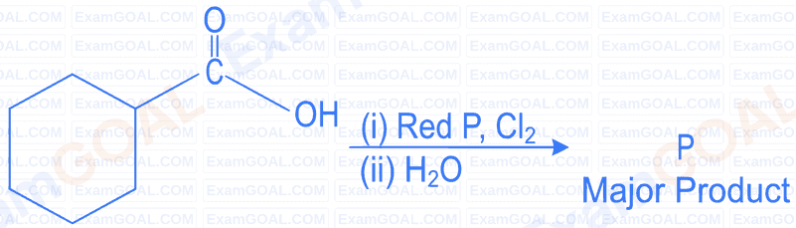
(A) $[Co(NH_3)_5Br]SO_4$ **(B)** $[Co(NH_3)_5SO_4]Br$ **(C)** $[Pt(NH_3)_4Cl_2]Br_2$ **(D)** $[Pt(NH_3)_4Br_2]Cl_2$

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Q9. MCQ (Single Correct Answer)

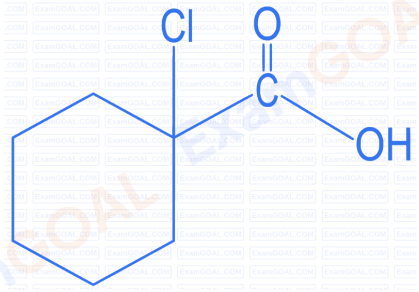
Open solution

Consider the given chemical reaction

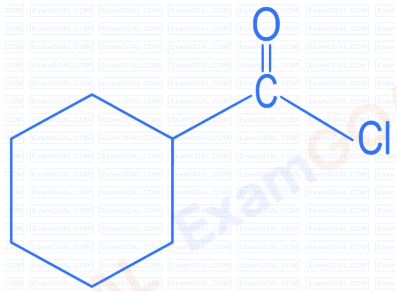


Identify the product P.

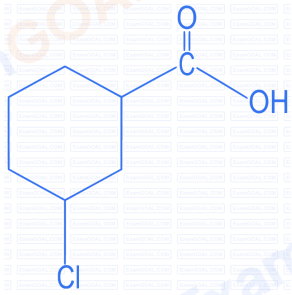
(A)



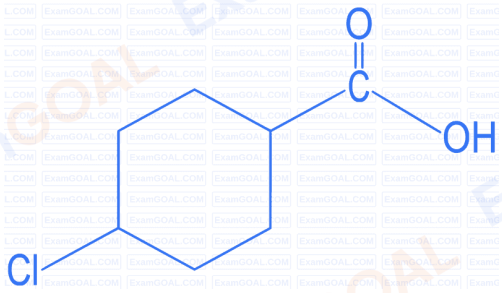
(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)

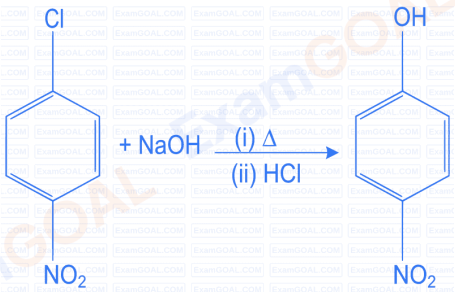
Open solution

Choose the reaction which is not possible:

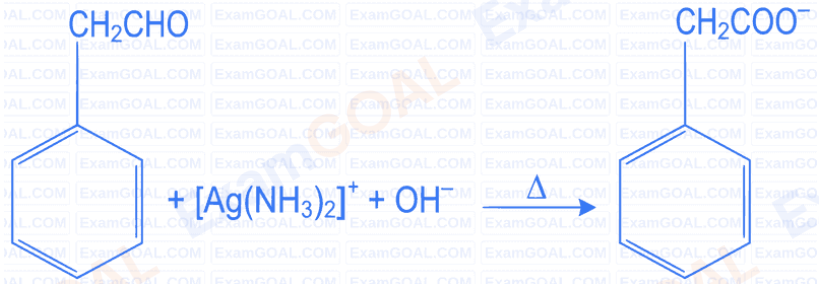
(A)



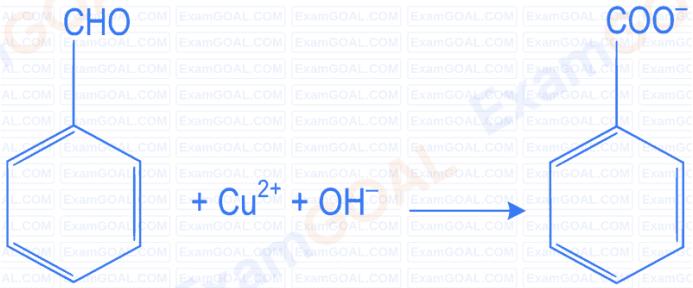
(B)



(C)



(D)

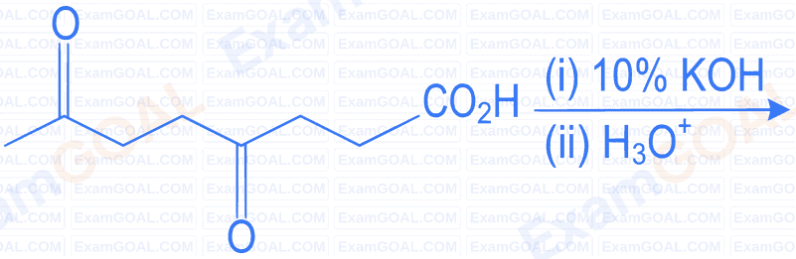


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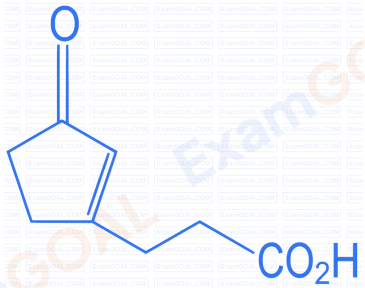
Q11. MCQ (Single Correct Answer)

Open solution

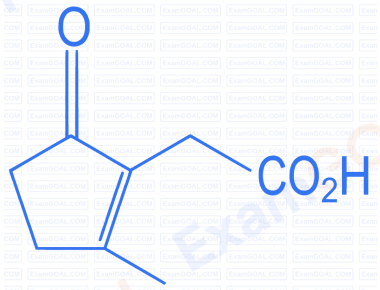
Which among the following will be the major product of the given reaction?



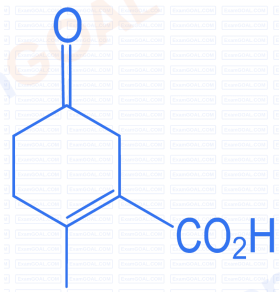
(A)



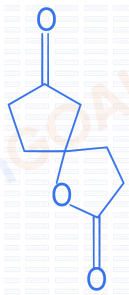
(B)



(C)



(D)



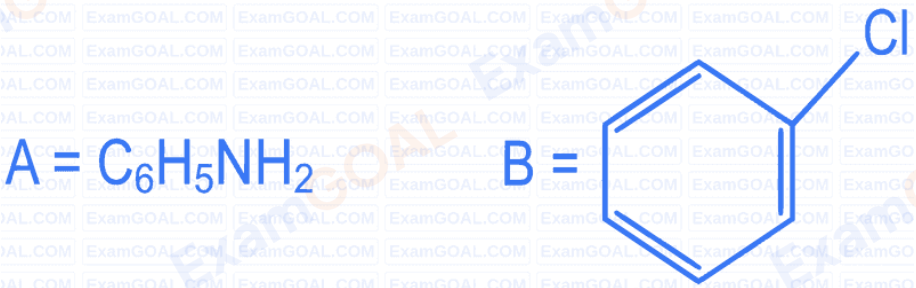
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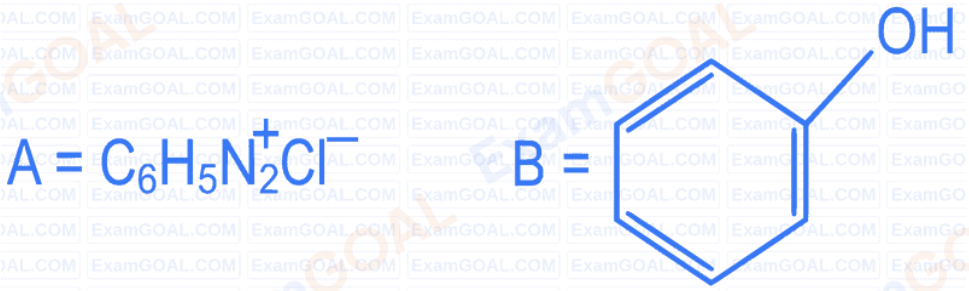
Q12. MCQ (Single Correct Answer)

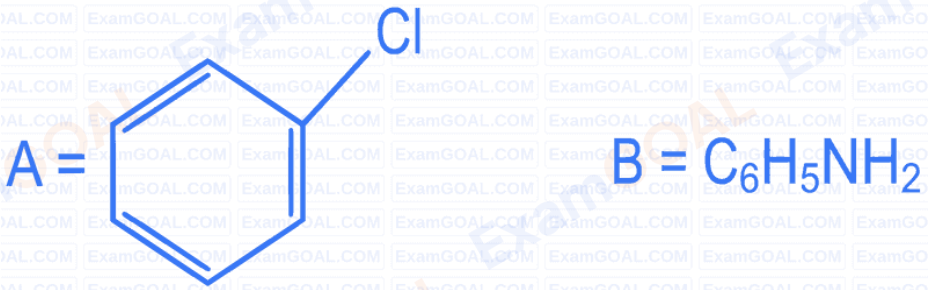


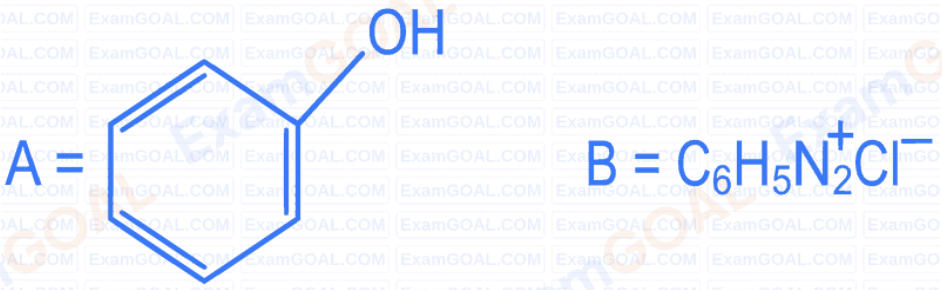
Open solution

Consider the above reaction sequence. Identify the component A and component B :

(A) 

(B) 

(C) 

(D) 

Q13. MCQ (Single Correct Answer)

The sugar produced after complete hydrolysis of DNA is

(A) a pentose sugar.
(B) a hexose sugar.
(C) a polysaccharide.
(D) a disaccharide.

Open solution

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Q14. MCQ (Single Correct Answer)

The reagent neutral ferric chloride is used to detect the presence of _____

(A) sulphide ion and alcoholic –OH group.

(B) acetate ion and phenolic –OH group.

(C) sulphide ion and phenolic –OH group.

(D) acetate ion and alcoholic –OH group.

Open solution

Q15. Numerical

Blister copper is produced by reaction of copper oxide with copper sulphide.

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$

When 2.86×10^3 g of Cu_2O and 4.77×10^3 g of Cu_2S are used for reaction, the mass of copper produced is _____ g. (nearest integer)

(Atomic mass of Cu = 63.5 a.m. u, S = 32.0 a.m. u, O = 16.0 a.m. u)

Open solution

Q16. Numerical

Amongst the following, the number of molecule/(s) having net resultant dipole moment is _____.

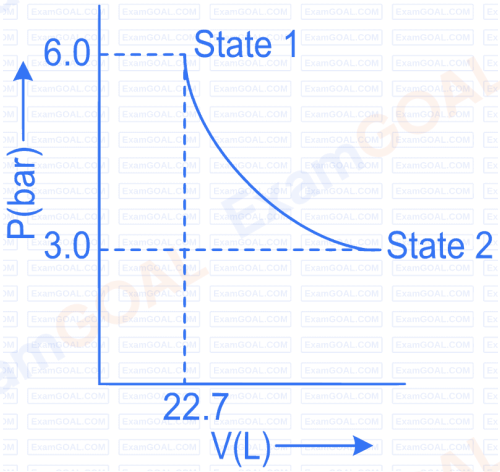
NF_3, BF_3, BeF_2, $CHCl_3$, H_2S, SiF_4, CCl_4, PF_5

Open solution

Q17. Numerical

1.0 mol of monoatomic ideal gas is expanded from state 1 to state 2 as shown in the figure. The magnitude of the work done for the expansion of gas from state 1 to state 2 at 300 K is _____ J. (Nearest integer)

(Given : R = 8.3 J K^{-1} mol^{-1}, ln10 = 2.3, log2 = 0.30)



Open solution

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Q18. Numerical

For the reaction P $\rightarrow$ B, the values of frequency factor A and activation energy E_A are 4×10^{13} s^{-1} and 8.3 kJ mol^{-1} respectively. If the reaction is of first order, the temperature at which the rate constant is 2×10^{-6} s^{-1} is _____ $\times 10^{-1}$ K.

(Given : ln 10 = 2.3, R = 8.3 J K^{-1} mol^{-1}, log2 = 0.30)

Open solution

Q19. Numerical

In the following brown complex, the oxidation state of iron is +_____.

$$[Fe(H_2O)_6]^{2+} + NO \rightarrow \underset{\text{Brown complex}}{[Fe(H_2O)_5(NO)]^{2+}} + H_2O$$

Open solution

Q20. Numerical

Spin only magnetic moment (μ_s) of $K_3[Fe(CN)_6]$ is _____ B.M.

(Nearest integer)

Open solution

Q21. Numerical

An organic compound with 51.6% sulfur is heated in a Carius tube. The amount of this compound which will form 0.752 g of barium sulphate is _____ $\times 10^{-1}$ g.

(Given molar mass of barium sulphate 233 g mol^{-1}) (Nearest integer).

Open solution

Q22. Numerical

A hydrocarbon 'X' is found to have molar mass of 80. A 10.0 mg of compound 'X' on hydrogenation consumed 8.40 mL of H_2 gas (measured at STP). Ozonolysis of compound 'X' yields only formaldehyde and dialdehyde. The total number of fragments/molecules produced from the ozonolysis of compound 'X' is _____.

Open solution

Mathematics 19 questions

Q23. MCQ (Single Correct Answer)

Let $S_1 = \left\{x \in R - \{1,2\} : \frac{(x+2)(x^2+3x+5)}{-2+3x-x^2} \geq 0\right\}$ and $S_2 = \left\{x \in R : 3^{2x} - 3^{x+1} - 3^{x+2} + 27 \leq 0\right\}$. Then, $S_1 \cup S_2$ is equal to :

(A) $(-\infty, -2] \cup (1, 2)$ (B) $(-\infty, -2] \cup [1, 2]$ (C) $(-2, 1] \cup [2, \infty)$ (D) $(-\infty, 2]$

Open solution

Q24. MCQ (Single Correct Answer)

The real part of the complex number $\frac{(1+2i)^8 \cdot (1-2i)^2}{(3+2i) \cdot \overline{(4-6i)}}$ is equal to :

(A) $\frac{500}{13}$ (B) $\frac{110}{13}$ (C) $\frac{55}{6}$ (D) $\frac{550}{13}$

Open solution

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Q25. MCQ (Single Correct Answer)

Let S be the set of all integral values of α for which the sum of squares of two real roots of the quadratic equation $3x^2 + (\alpha - 6)x + (\alpha + 3) = 0$ is minimum. Then S :

(A) is an empty set

(B) is a singleton

(C) contains exactly two elements

(D) contains more than two elements

Open solution

Q26. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 1 & -2 & \alpha \\ \alpha & 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & \alpha \\ -1 & 2 \\ 4 & -5 \end{bmatrix}$, $\alpha \in C$. Then the absolute value of the sum of all values of α for which det(AB) = 0 is :

(A) 3

(B) 4

(C) 2

(D) 5

Open solution

Q27. MCQ (Single Correct Answer)

For two positive real numbers a and b such that $\frac{1}{a^2} + \frac{1}{b^3} = 4$, then minimum value of the constant term in the expansion of $\left(ax^{\frac{1}{8}} + bx^{-\frac{1}{12}}\right)^{10}$ is :

(A) $\frac{105}{2}$

(B) $\frac{105}{4}$

(C) $\frac{105}{8}$

(D) $\frac{105}{16}$

Open solution

Q28. MCQ (Single Correct Answer)

If xy^4 attains maximum value at the point (x, y) on the line passing through the points (50 + α, 0) and (0, 50 + α), α > 0, then (x, y) also lies on the line :

(A) y = 4x

(B) x = 4y

(C) y = 4x + α

(D) x = 4y $-\ \alpha$

Open solution

Q29. MCQ (Single Correct Answer)

Let $f(x) = 4x^3 - 11x^2 + 8x - 5$, $x \in R$. Then f :

(A) has a local minima at $x = \frac{1}{2}$

(B) has a local minima at $x = \frac{3}{4}$

(C) is increasing in $\left(\frac{1}{2}, \frac{3}{4}\right)$

(D) is decreasing in $\left(\frac{1}{2}, \frac{4}{3}\right)$

Open solution

Q30. MCQ (Single Correct Answer)

Let m and M respectively be the minimum and the maximum values of $f(x) = \sin^{-1}2x + \sin 2x + \cos^{-1}2x + \cos 2x$, $x \in \left[0, \frac{\pi}{8}\right]$. Then m + M is equal to :

(A) $1 + \sqrt{2} + \pi$

(B) $\left(1 + \sqrt{2}\right)\pi$

(C) $\pi + \sqrt{2}$

(D) $1 + \pi$

Open solution

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Q31. MCQ (Single Correct Answer)

Let α_1, α_2 ($\alpha_1 < \alpha_2$) be the values of α for the points $(\alpha, -3)$, $(2, 0)$ and $(1, \alpha)$ to be collinear. Then the equation of the line, passing through (α_1, α_2) and making an angle of $\frac{\pi}{3}$ with the positive direction of the x-axis, is :

(A) $x - \sqrt{3}y - 3\sqrt{3} + 1 = 0$

(B) $\sqrt{3}x - y + \sqrt{3} + 3 = 0$

(C) $x - \sqrt{3}y + 3\sqrt{3} + 1 = 0$

(D) $\sqrt{3}x - y + \sqrt{3} - 3 = 0$

Open solution

Q32. MCQ (Single Correct Answer)

Let the eccentricity of the ellipse $x^2 + a^2y^2 = 25a^2$ be b times the eccentricity of the hyperbola $x^2 - a^2y^2 = 5$, where a is the minimum distance between the curves y = e^x and y = $\log_e x$. Then $a^2 + \frac{1}{b^2}$ is equal to :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) 3 (D) 5

Open solution

Q33. MCQ (Single Correct Answer)

Let $\alpha = \tan\left(\frac{5\pi}{16}\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)\right)$ and $\beta = \cos\left(\sin^{-1}\left(\frac{4}{5}\right) + \sec^{-1}\left(\frac{5}{3}\right)\right)$ where the inverse trigonometric functions take principal values. Then, the equation whose roots are α and β is :

(A) $15x^2 - 8x - 7 = 0$

(B) $5x^2 - 12x + 7 = 0$

(C) $25x^2 - 18x - 7 = 0$

(D) $25x^2 - 32x + 7 = 0$

Open solution

Q34. Numerical

The number of 6-digit numbers made by using the digits 1, 2, 3, 4, 5, 6, 7, without repetition and which are multiple of 15 is _____.

Open solution

Q35. Numerical

Suppose $\lim\limits_{x\to 0}\frac{F(x)}{x^3}$ exists and is equal to L, where

$$F(x) = \begin{vmatrix} a + \sin\frac{x}{2} & -b\cos x & 0 \\ -b\cos x & 0 & a + \sin\frac{x}{2} \\ 0 & a + \sin\frac{x}{2} & -b\cos x \end{vmatrix}.$$

Then, -112 L is equal to _____.

Open solution

Q36. Numerical

If for some $\alpha > 0$, the area of the region $\{(x, y) : |x + \alpha| \le y \le 2 - |x|\}$ is equal to $\frac{3}{2}$, then the area of the region $\{(x, y) : 0 \le y \le x + 2\alpha,\ |x| \le 1\}$ is equal to _____.

Open solution

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Q37. Numerical

Let $f(t) = \int_0^t e^{x^3}\left(\frac{x^8}{(x^6+2x^3+2)^2}\right)dx$. If $f(1) + f'(1) = \alpha e - \frac{1}{6}$, then the value of 150α is equal to _____.

Open solution

Q38. Numerical

A hostel has 100 students. On a certain day (consider it day zero) it was found that two students are infected with some virus. Assume that the rate at which the virus spreads is directly proportional to the product of the number of infected students and the number of non-infected students. If the number of infected students on 4th day is 30, then number of infected students on 8th day will be _____.

Open solution

Q39. Numerical

Let PQ be a focal chord of length 6.25 units of the parabola $y^2 = 4x$. If O is the vertex of the parabola, then 10 times the area (in sq. units) of ΔPOQ is equal to _____.

Open solution

Q40. Numerical

Consider a triangle ABC whose vertices are A(0, α, α), B(α, 0, α) and C(α, α, 0), $\alpha > 0$. Let D be a point moving on the line $x + z - 3 = 0 = y$ and G be the centroid of ΔABC. If the minimum length of GD is $\sqrt{\frac{57}{2}}$, then α is equal to _____.

Open solution

Q41. Numerical

The probability distribution of X is :

X	0	1	2	3
P(X)	$\frac{1-d}{4}$	$\frac{1+2d}{4}$	$\frac{1-4d}{4}$	$\frac{1+3d}{4}$

For the minimum possible value of d, sixty times the mean of X is equal to _____.

Open solution

Physics 25 questions

Q42. MCQ (Single Correct Answer)

At t = 0, truck, starting from rest, moves in the positive x-direction at uniform acceleration of 5 ms^{-2}. At t = 20 s, a ball is released from the top of the truck. The ball strikes the ground in 1 s after the release. The velocity of the ball, when it strikes the ground, will be :

(Given g = 10 ms^{-2})

(A) $100\hat{i} - 10\hat{j}$ (B) $10\hat{i} - 100\hat{j}$ (C) $100\hat{i}$ (D) $-10\hat{j}$

Open solution

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Q43. MCQ (Single Correct Answer)

If n main scale divisions coincide with (n + 1) vernier scale divisions. The least count of vernier callipers, when each centimetre on the main scale is divided into five equal parts, will be :

(A) $\frac{2}{n+1}$ mm (B) $\frac{5}{n+1}$ mm (C) $\frac{1}{2n}$ mm (D) $\frac{1}{5n}$ mm

Open solution

Q44. MCQ (Single Correct Answer)

The radii of two planets A and B are in the ratio 2 : 3. Their densities are 3ρ and 5ρ respectively. The ratio of their acceleration due to gravity is :

(A) 9 : 4 (B) 9 : 8 (C) 9 : 10 (D) 2 : 5

Open solution

Q45. MCQ (Single Correct Answer)

Two projectiles P_1 and P_2 thrown with speed in the ratio $\sqrt{3} : \sqrt{2}$, attain the same height during their motion. If P_2 is thrown at an angle of 60° with the horizontal, the angle of projection of P_1 with horizontal will be :

(A) 15° (B) 30° (C) 45° (D) 60°

Open solution

Q46. MCQ (Single Correct Answer)

An air bubble of negligible weight having radius r rises steadily through a solution of density σ at speed v. The coefficient of viscosity of the solution is given by :

(A) $\eta = \frac{4r\sigma g}{9v}$ (B) $\eta = \frac{2r^2\sigma g}{9v}$

(C) $\eta = \frac{2\pi r^2\sigma g}{9v}$ (D) $\eta = \frac{2r^2\sigma g}{3\pi v}$

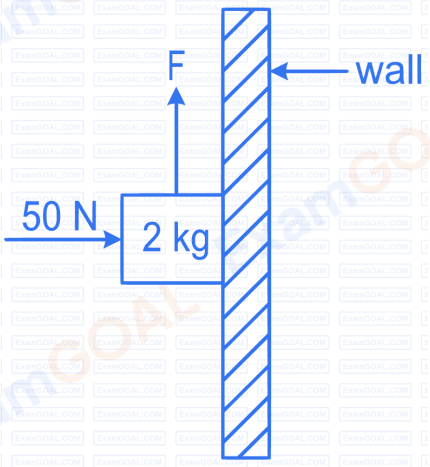
Open solution

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Q47. MCQ (Single Correct Answer)

A 2 kg block is pushed against a vertical wall by applying a horizontal force of 50 N. The coefficient of static friction between the block and the wall is 0.5. A force F is also applied on the block vertically upward (as shown in figure). The maximum value of F applied, so that the block does not move upward, will be :

(Given : g = 10 ms^{-2})



Open solution

(A) 10 N (B) 20 N (C) 25 N (D) 45 N

Q48. MCQ (Single Correct Answer)

Two bodies A and B of masses 5 kg and 8 kg are moving such that the momentum of body B is twice that of the body A. The ratio of their kinetic energies will be :

Open solution

(A) 4 : 5 (B) 2 : 5 (C) 5 : 4 (D) 5 : 2

Q49. MCQ (Single Correct Answer)

The pressure of the gas in a constant volume gas thermometer is 100 cm of mercury when placed in melting ice at 1 atm. When the bulb is placed in a liquid, the pressure becomes 180 cm of mercury. Temperature of the liquid is :

(Given 0°C = 273 K)

Open solution

(A) 300 K (B) 400 K (C) 600 K (D) 491 K

Q50. MCQ (Single Correct Answer)

A coil of n number of turns wound tightly in the form of a spiral with inner and outer radii r_1 and r_2 respectively. When a current of strength I is passed through the coil, the magnetic field at its centre will be :

Open solution

(A) $\frac{\mu_0 nI}{2(r_2-r_1)}$ (B) $\frac{\mu_0 nI}{r_2}$

(C) $\frac{\mu_0 nI}{r_2-r_1}\log_e\frac{r_1}{r_2}$ (D) $\frac{\mu_0 nI}{2(r_2-r_1)}\log_e\frac{r_2}{r_1}$

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Q51. MCQ (Single Correct Answer)

Open solution

C_o is the capacitance of a parallel plate capacitor with air as a medium between the plates (as shown in Fig. 1). If half space between the plates is filled with a dielectric of relative permittivity ε_r (as shown in Fig. 2), the new capacitance of the capacitor will be :



Fig. 1



Fig. 2

(A) $\frac{C_o}{2}(1+\varepsilon_r)$

(B) $C_o+\varepsilon_r$

(C) $\frac{C_o\varepsilon_r}{2}$

(D) $C_o(1+\varepsilon_r)$

Q52. MCQ (Single Correct Answer)

Open solution

A sample of monoatomic gas is taken at initial pressure of 75 kPa. The volume of the gas is then compressed from 1200 cm^3 to 150 cm^3 adiabatically. In this process, the value of workdone on the gas will be :

(A) 79 J

(B) 405 J

(C) 4050 J

(D) 9590 J

Q53. MCQ (Single Correct Answer)

Open solution

Which of the following equations correctly represents a travelling wave having wavelength λ = 4.0 cm, frequency ν = 100 Hz and travelling in positive x-axis direction?

(A) $y = A\sin[(0.50\,\pi\,cm^{-1})x - (100\,\pi\,s^{-1})t]$

(B) $y = A\sin\ 2\pi[(0.25\ cm^{-1})x - (50\,s^{-1})t]$

(C) $y = A\sin\left[\left(\frac{2\pi}{4}\ cm^{-1}\right)x - \left(\frac{2\pi}{100}\ s^{-1}\right)t\right]$

(D) $y = A\sin\ \pi[(0.5\ cm^{-1})x - (200\ s^{-1})t]$

Q54. MCQ (Single Correct Answer)

Open solution

An expression for oscillating electric field in a plane electromagnetic wave is given as E_z = 300 $\sin(5\pi \times 10^3 x - 3\pi \times 10^{11} t)$ Vm^{-1}

Then, the value of magnetic field amplitude will be :

(Given : speed of light in Vacuum c = 3×10^8 ms^{-1})

(A) 1×10^{-6} T

(B) 5×10^{-6} T

(C) 18×10^9 T

(D) 21×10^9 T

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Q55. MCQ (Single Correct Answer)

An electric cable of copper has just one wire of radius 9 mm. Its resistance is 14 Ω. If this single copper wire of the cable is replaced by seven identical well insulated copper wires each of radius 3 mm connected in parallel, then the new resistance of the combination will be :

Open solution

(A) 9 Ω (B) 18 Ω (C) 28 Ω (D) 126 Ω

Q56. MCQ (Single Correct Answer)

Find the ratio of maximum intensity to the minimum intensity in the interference pattern if the widths of the two slits in Young's experiment are in the ratio of 9 : 16. (Assuming intensity of light is directly proportional to the width of slits)

Open solution

(A) 3 : 4 (B) 4 : 3 (C) 7 : 1 (D) 49 : 1

Q57. MCQ (Single Correct Answer)

A source of monochromatic light liberates 9×10^{20} photon per second with wavelength 600 nm when operated at 400 W. The number of photons emitted per second with wavelength of 800 nm by the source of monochromatic light operating at same power will be :

Open solution

(A) 12×10^{20} (B) 6×10^{20} (C) 9×10^{20} (D) 24×10^{20}

Q58. MCQ (Single Correct Answer)

A hydrogen atom in ground state absorbs 12.09 eV of energy. The orbital angular momentum of the electron is increased by :

Open solution

(A) 1.05×10^{-34} Js (B) 2.11×10^{-34} Js

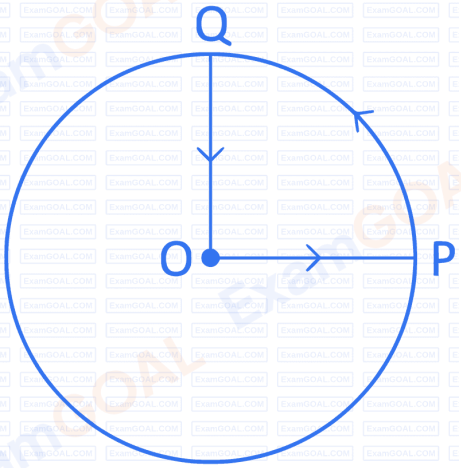
(C) 3.16×10^{-34} Js (D) 4.22×10^{-34} Js

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Q59. Numerical

A person starts his journey from centre 'O' of the park and comes back to the same position following path OPQO as shown in the figure. The radius of path taken by the person is 200 m and he takes 3 min 58 sec to complete his journey. The average speed of the person is _____ ms^{-1}. (take π = 3.14)

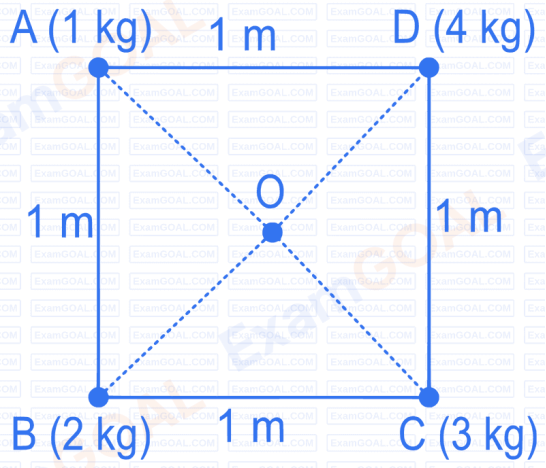
Open solution



Q60. Numerical

Four particles with a mass of 1 kg, 2 kg, 3 kg and 4 kg are situated at the corners of a square with side 1 m (as shown in the figure). The moment of inertia of the system, about an axis passing through the point O and perpendicular to the plane of the square, is _____ kg m^2.

Open solution



Q61. Numerical

The excess pressure inside a liquid drop is 500 Nm^{-2}. If the radius of the drop is 2 mm, the surface tension of liquid is $x \times 10^{-3}$ Nm^{-1}. The value of x is _____.

Open solution

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Q62. Numerical

Eight similar drops of mercury are maintained at 12 V each. All these spherical drops combine into a single big drop. The potential energy of bigger drop will be _____ E. Where E is the potential energy of a single smaller drop.

Open solution

Q63. Numerical

A series LCR circuit with $R = \frac{250}{11}\ \Omega$ and $X_L = \frac{70}{11}\ \Omega$ is connected across a 220 V, 50 Hz supply. The value of capacitance needed to maximize the average power of the circuit will be _____ μF. (Take : $\pi = \frac{22}{7}$)

Open solution

Q64. Numerical

The refractive index of an equilateral prism is $\sqrt{2}$. The angle of emergence under minimum deviation position of prism, in degree, is _____.

Open solution

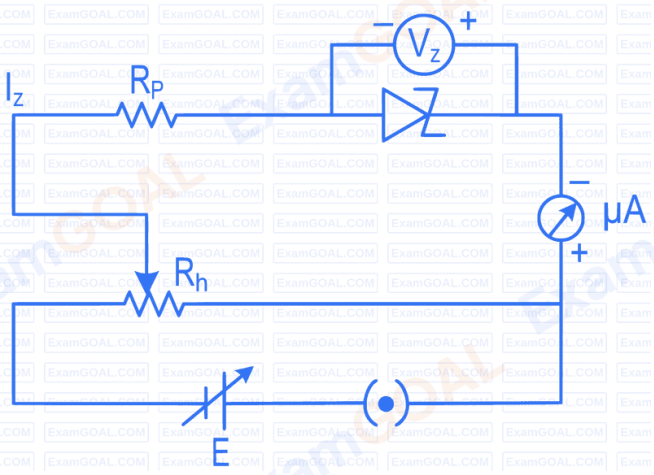
Q65. Numerical

A hydrogen atom in its first excited state absorbs a photon of energy x $\times$ 10^{-2} eV and excited to a higher energy state where the potential energy of electron is -1.08 eV. The value of x is _____.

Open solution

Q66. Numerical

The circuit diagram used to study the characteristic curve of a zener diode is connected to variable power supply (0 $-$ 15 V) as shown in figure. A zener diode with maximum potential V_z = 10 V and maximum power dissipation of 0.4 W is connected across a potential divider arrangement. The value of resistance R_P connected in series with the zener diode to protect it from the damage is _____ Ω.



Open solution

Out of syllabus 24 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q67. MCQ (Single Correct Answer) · **Chemistry**

An element X has a body centred cubic (bcc) structure with a cell edge of 200 pm. The density of the element is 5 g cm^{-3}. The number of atoms present in 300 g of the element X is _____.

Given : Avogadro constant, $N_A = 6.0 \times 10^{23}$ mol^{-1}.

Open solution

(A) 5 N_A (B) 6 N_A (C) 15 N_A (D) 25 N_A

Q68. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

	List-I		List-II
(a)	Sodium hydride	(i)	Lewis acid
(b)	Silane	(ii)	Saline hydride
(c)	Vanadium hydride	(iii)	Molecular hydride
(d)	Aluminium hydride	(iv)	Non-stoichiometric hydride

Open solution

Correct answer is :

(A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(B) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(C) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
(D) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

Q69. MCQ (Single Correct Answer) · **Chemistry**

Correct statement about alkali metal oxides is :

Open solution

(A) peroxides are colored.
(B) superoxides are paramagnetic.
(C) oxides are paramagnetic.
(D) peroxides are both colored and paramagnetic.

Q70. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following chemicals has not been used for the dry cleaning of clothes?

Open solution

(A) Tetrachloroethene
(B) Liquid CO_2
(C) Ethanol
(D) H_2O_2

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Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

TLC analysis of a mixture having 3 components (A, B, C) using silica gel as the stationary phase gave following R_f values; for A = 0.72, B = 0.48, C = 0.20.

Regarding the above observations, which one of the following statements is not correct for column chromatography of the mixture?

(A) A is the most non-polar component

(B) C is the most soluble component in mobile phase

(C) A will be eluted first

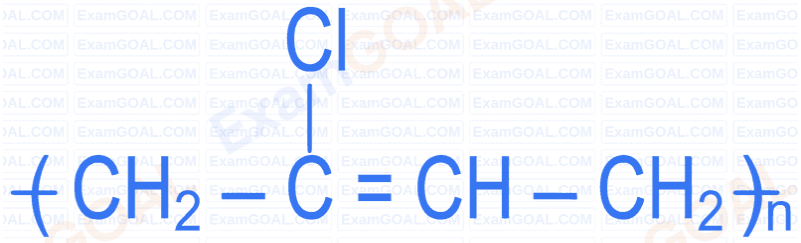
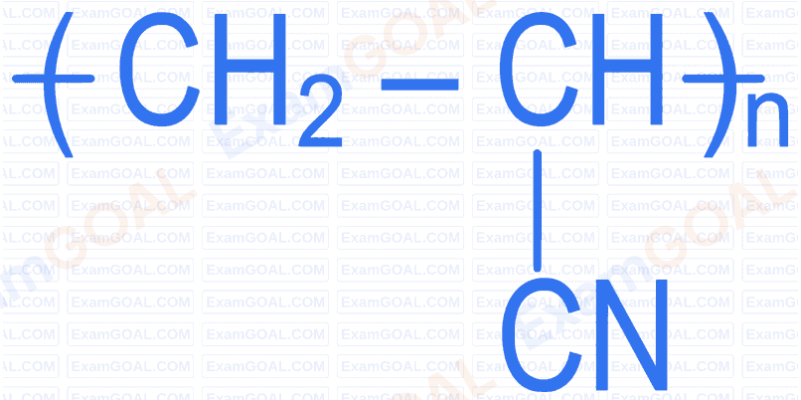
(D) C is the most strongly absorbed component

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II.

	List-I Polymer		List-II Repeat Unit
(a)	Acrilan	(i)	
(b)	Neoprene	(ii)	
(c)	Polystyrene	(iii)	
(d)	Buna-N	(iv)	

Choose the correct match from the options given below :

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(B) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(C) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(D) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

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Q73. Numerical · **Chemistry**

100 mL of 0.3 M acetic acid is shaken with 0.8 g of wood charcoal. The final concentration of acetic acid in the solution after adsorption is 0.125 M. The mass of acetic acid adsorbed per gram of carbon is _____ $\times 10^{-4}$ g.

(Given : Molar mass of acetic acid = 60 g mol^{-1})

Open solution

Q74. Numerical · **Chemistry**

How many of the following drugs belong to the tranquilizer class _____ ?

Veronal, Luminal, Propanal, Seconal

Open solution

Q75. MCQ (Single Correct Answer) · **Mathematics**

Let A and B be two square matrices of order 2. If $det\,(A) = 2$, $det\,(B) = 3$ and $\det\left((\det\,(5(\det\,A)B))A^2\right) = 2^a3^b5^c$ for some a, b, c, $\in$ N, then a + b + c is equal to :

(A) 10 (B) 12 (C) 13 (D) 14

Open solution

Q76. MCQ (Single Correct Answer) · **Mathematics**

The value of $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots. + \frac{1}{1+2+3+\ldots..+11}$ is equal to:

(A) $\frac{20}{11}$ (B) $\frac{11}{6}$ (C) $\frac{241}{132}$ (D) $\frac{21}{11}$

Open solution

Q77. MCQ (Single Correct Answer) · **Mathematics**

$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{2r^2-7rn+6n^2}$ is equal to :

(A) $\log_e\left(\frac{\sqrt{3}}{2}\right)$ (B) $\log_e\left(\frac{3\sqrt{3}}{4}\right)$

(C) $\log_e\left(\frac{27}{4}\right)$ (D) $\log_e\left(\frac{4}{3}\right)$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, $a, b, c \in R$, represents a circle with center (α, β). Then, $\alpha + 2\beta$ is equal to :

(A) -1 (B) 0 (C) 1 (D) 2

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

Consider three circles:

$C_1 : x^2 + y^2 = r^2$

$C_2 : (x-1)^2 + (y-1)^2 = r^2$

$C_3 : (x-2)^2 + (y-1)^2 = r^2$

If a line L : y = mx + c be a common tangent to C_1, C_2 and C_3 such that C_1 and C_3 lie on one side of line L while C_2 lies on other side, then the value of $20(r^2 + c)$ is equal to :

(A) 23 (B) 15 (C) 12 (D) 6

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point (3, 2, -1) from the plane $3x - y + 4z + 1 = 0$ along the line $\frac{2-x}{2} = \frac{y-3}{2} = \frac{z+1}{1}$ is equal to :

(A) 9 (B) 6 (C) 3 (D) 2

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let a vector $\vec{c}$ be coplanar with the vectors $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. If the vector $\vec{c}$ also satisfies the conditions $\vec{c}.\left[\left(\vec{a}+\vec{b}\right) \times \left(\vec{a} \times \vec{b}\right)\right] = -42$ and $\left(\vec{c} \times \left(\vec{a} - \vec{b}\right)\right).\hat{k} = 3$, then the value of $|\vec{c}|^2$ is equal to :

(A) 24 (B) 29 (C) 35 (D) 42

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If a random variable X follows the Binomial distribution B(5, p) such that P(X = 0) = P(X = 1), then $\frac{P(X=2)}{P(X=3)}$ is equal to :

(A) 1 (B) 10 (C) 25 (D) 5

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The conditional statement

$((p \wedge q) \rightarrow ((\sim p) \vee r)) \vee (((\sim p) \vee r) \rightarrow (p \wedge q))$ is :

(A) a tautology (B) a contradiction

(C) equivalent to $p \wedge q$ (D) equivalent to $(\sim p) \vee r$

Open solution

Q84. Numerical · **Mathematics**

Let for $f(x) = a_0x^2 + a_1x + a_2$, $f'(0) = 1$ and $f'(1) = 0$. If a_0, a_1, a_2 are in an arithmatico-geometric progression, whose corresponding A.P. has common difference 1 and corresponding G.P. has common ratio 2, then f(4) is equal to _____.

Open solution

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Q85. Numerical · **Mathematics**

Let $S_1 = \{x \in [0, 12\pi] : \sin^5 x + \cos^5 x = 1\}$

and $S_2 = \{x \in [0, 8\pi] : \sin^7 x + \cos^7 x = 1\}$

Then $n(S_1) - n(S_2)$ is equal to _____.

Open solution

Q86. MCQ (Single Correct Answer) · **Physics**

A cyclotron is working at a frequency of 10 MHz. If the radius of its dees is 60 cm. The maximum kinetic energy of accelerated proton will be :

(Take : e = 1.6 $\times$ 10^{-19} C, m_p = 1.67 $\times$ 10^{-27} kg)

(A) 7.4 MeV (B) 14.86 MeV (C) 7.4 GeV (D) 704 GeV

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

In series RLC resonator, if the self inductance and capacitance become double, the new resonant frequency (f_2) and new quality factor (Q_2) will be :

(f_1 = original resonant frequency, Q_1 = original quality factor)

(A) $f_2 = \frac{f_1}{2}$ and $Q_2 = Q_1$ (B) $f_2 = f_1$ and $Q_2 = \frac{Q_1}{Q_2}$

(C) $f_2 = 2f_1$ and $Q_2 = Q_1$ (D) $f_2 = f_1$ and $Q_2 = 2Q_1$

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

A speech signal given by 11 sin(2200 πt) V is used for amplitude modulation with a carrier signal given by 44 sin(6600 πt) V. The minimum amplitude of modulated wave will be :

(A) 33 V (B) 55 V (C) 8.25 V (D) 13.75 V

Open solution

Q89. Numerical · **Physics**

An employee of a factory moving away from his workplace by a car listens to the siren of the factory. He drives the car at the speed of 72 kmh^{-1} in the direction of wind which is blowing at 72 kmh^{-1} speed. Frequency of siren is 720 Hz. The employee hears an apparent frequency of _____ Hz.

(Assume speed of sound to be 340 ms^{-1})

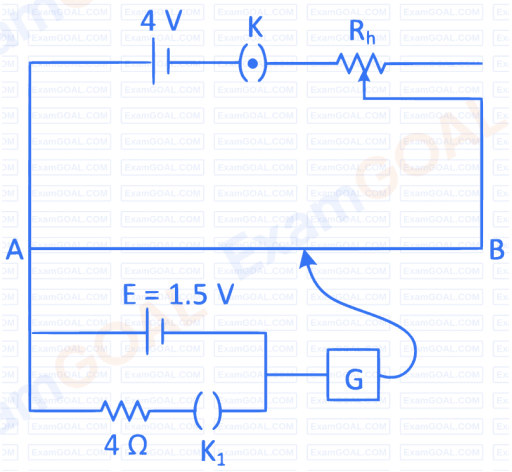
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Q90. Numerical · **Physics**

Open solution

The circuit diagram of potentiometer used to measure the internal resistance of a cell (E) is shown in figure. The key 'K' is kept closed so as to send constant current through potentiometer wire. When key 'K_1' is kept open the null point is found to be at 120 cm on the potentiometer wire. When the key 'K_1' is closed the null point is shifted at 80 cm at the potentiometer wire. The internal resistance of the given cell is _____ Ω.



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Answer Key

Chemistry

1. **A**	2. **A**	3. **C**	4. **B**	5. **C**
6. **A**	7. **D**	8. **A**	9. **A**	10. **D**
11. **B**	12. **B**	13. **A**	14. **B**	15. **3810**
16. **3**	17. **1718**	18. **225**	19. **1**	20. **2**
21. **2**	22. **4**			

Mathematics

23. **B**	24. **D**	25. **A**	26. **A**	27. **C**
28. **A**	29. **D**	30. **A**	31. **B**	32. **D**
33. **C**	34. **360**	35. **14**	36. **4**	37. **16**
38. **90**	39. **25**	40. **6**	41. **75**	

Physics

42. **A**	43. **A**	44. **D**	45. **C**	46. **B**
47. **D**	48. **B**	49. **D**	50. **D**	51. **A**
52. **B**	53. **D**	54. **B**	55. **B**	56. **D**
57. **A**	58. **B**	59. **3**	60. **5**	61. **500**
62. **32**	63. **500**	64. **45**	65. **286**	66. **125**

Out of syllabus

67. **D**	68. **A**	69. **B**	70. **C**	71. **B**
72. **C**	73. **13125**	74. **3**	75. **B**	76. **B**
77. **B**	78. **C**	79. **D**	80. **C**	81. **C**
82. **D**	83. **A**	84. **2**	85. **4**	86. **A**
87. **A**	88. **A**	89. **680**	90. **2**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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